From a practical perspective, there is also tooling available, such as the SLI Analyzer (https://docs.nobl9.com/slocademy/SLI_Analyzer/) by Nobl9, enabling you to generate SLOs based on (historical) SLIs. Now that we have an idea what SLOs are (goals) and how to measure to what extent we meet them (SLIs), let's move up the stack to the contract between the service provider and the service customer.

10.1.4 Service level agreement

An SLA is what you, as a service provider, promise to your (service) customer, which is mostly a contractual topic. A more tongue in cheek way to say that is, "SLOs are written in engineer, and SLAs are written in lawyer" (kudos to Jamie!). This is mostly between the sales folks and the customers, based on and informed by the SLO that product and engineering commit to.

For example, if product and engineering commit to an SLO that states that an API is available for 99% of the quarter and, after a quarter, you were only able to make the API available for, say, 98% of the time, then you have violated the availability SLA. This may result in consequences, such as a refund or even paying a penalty, depending on the type of service you operate or kind of relationship you have with the customer.

10.2 Implementing SLOs

Now that you know about the terminology, let us discuss how to go about implementing hypothetical SLAs, SLOs, and SLIs. We will start off with an example conversation between sales, product, and engineering and then have a look at a concrete example of how to implement SLOs within the Prometheus ecosystem.

10.2.1 High-level example

Imagine the following: your service team plans to launch a new API for a conversational chat bot called Chat2Much. The hypothetical conversation goes something like the following:

> *Sales → Product: Yo, I hear Chat2Much is soon ready to launch. I'm working on the SLA and wanted to understand what we can promise to customers about an API availability rate over the month as well as the request failure rate.*
>
> *Product → Sales: Sure thing, let me get back to you on this.*
>
> *Product → Engineering: I'd like to establish an SLO that sales can use as a basis for the Chat2Much SLA.*
>
> *Engineering → Product: What SLO did you have in mind for Chat2Much?*
>
> *Product → Engineering: I was thinking of committing to 99% API availability over a period of a month—what do you think? Also, can we do request count and errors?*
>
> *Engineering → Product: That sounds reasonable, I believe we can achieve 99.9% with the current head count and technology stack. We will implement the service metrics* `A`*,* `B`*, and* `C`*, which you can use as SLIs to track the SLO.*